

**K. J. Somaiya College of Engineering, Mumbai-77**  
(Autonomous College Affiliated to University of Mumbai)

**End Semester Exam**  
NOV-DEC 2021

**Max. Marks: 50**

**Duration: 1 Hr. 45 Min.**

Class: TY B.Tech

Name of the Course: Soft Computing

Course Code: **2UCE514**

Semester: V

Branch: Computer

**Instructions:**

- (1) **All questions are compulsory**
- (2) **Draw neat diagrams**
- (3) **Assume suitable data if necessary**

| Question No. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Max Marks |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Q1 (A)       | <p>1. What is back propagation?</p> <p>a) It is another name given to the curvy function in the perceptron</p> <p>b) It is the transmission of error back through the network to adjust the inputs</p> <p>c) It is the transmission of error back through the network to allow weights to be adjusted so that the network can learn.</p> <p>d) It is the transmission of error back through the network to adjust the outputs</p> <p>2. Which functions in Neural Networks are complex with many parameters.</p> <p>a) Linear Functions</p> <p>b) Nonlinear Functions</p> <p>c) Discrete Functions</p> <p>d) Exponential Functions</p> <p>3. The network that involves backward links from output to the input and hidden layers is called as</p> <p>a) Self organizing maps</p> <p>b) Perceptron's</p> <p>c) Recurrent neural network</p> <p>d) Multi layered perceptron</p> <p>4. In an Unsupervised learning</p> <p>a) Specific output values are given</p> <p>b) Specific output values are not given</p> <p>c) No specific Inputs are given</p> <p>d) Both inputs and outputs are given</p> <p>5. In artificial Neural Network interconnected processing elements are called</p> <p>a) nodes or neurons</p> <p>b) weights</p> <p>c) axons</p> <p>d) soma</p> | 10        |

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|        | <p>6. Who initiated the idea of Soft Computing</p> <ol style="list-style-type: none"> <li>Mc_Culloch</li> <li>Lofti A Zadeh</li> <li>Charles Darwin</li> <li>Rechenberg</li> </ol> <p>7. The region of universe that is characterized by complete membership in the set is called</p> <ol style="list-style-type: none"> <li>Core</li> <li>Support</li> <li>Boundary</li> <li>Fuzzy</li> </ol> <p>8. A fuzzy set whose membership function has at least one element <math>x</math> in the universe whose membership value is unity is called</p> <ol style="list-style-type: none"> <li>sub normal fuzzy sets</li> <li>normal fuzzy set</li> <li>convex fuzzy set</li> <li>concave fuzzy set</li> </ol> <p>9. ADALINE stands for</p> <ol style="list-style-type: none"> <li>Adaptive Linear Neural Element Network</li> <li>Adaptive Line Neural Network</li> <li>Adapt Line Neural Element Network</li> <li>Adaptive Linear Neural Network</li> </ol> <p>10. The store house of associated patterns which are encoded in some form</p> <ol style="list-style-type: none"> <li>Associative memory</li> <li>Commutative memory</li> <li>Neural networks</li> <li>Memory</li> </ol> |    |
| Q1 (B) | <p>Attempt any FIVE questions out of the following <b>(any 5 out of 7)</b></p> <ol style="list-style-type: none"> <li>Design two input AND logic using McCulloch Pitts Neuron model</li> <li>Explain linearly separable and non-linearly separable patterns with examples.</li> <li>Model the fuzzy set using suitable membership function. "Integer number considerably larger than 6"</li> <li>A neuron with 4 inputs has the weight vector <math>w = [1 \ 2 \ 3 \ 4]^t</math>. The activation function is linear that is, the activation function is given by <math>f(\text{net}) = 2 * \text{net}</math>. If the input vector is <math>X = [5 \ 6 \ 7 \ 8]^t</math>, then find the output of the neuron.</li> <li>Prove that first order derivative of unipolar continuous activation function is <math>f'(\text{net}) = O(1-O)</math> where <math>O</math> is output</li> <li>Explain any two types of fuzzy inference systems</li> <li>Explain with example perceptron learning rule.</li> </ol>                                                                                                                                                                            | 10 |

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| Q. 2 | <p>What is learning in Neural Network ? Compare different learning rules.</p> <p style="text-align: center;"><b>OR</b></p> <p>What is competitive learning and what are limitations of competitive learning.</p>                                                                                                                                      | 10 |
| Q. 3 | <p>Design a fuzzy controller for train approaching or leaving a station, the inputs are distance from a station and speed of the train. The output is brake power used.</p> <ol style="list-style-type: none"> <li>i. Use triangular membership function.</li> <li>ii. Four descriptors for each variables</li> <li>iii. Five to six rules</li> </ol> | 10 |
| Q. 4 | <p>Write short note on (any two)</p> <ol style="list-style-type: none"> <li>i. Character recognition using neural networks</li> <li>ii. Kohonen 's self-organizing network</li> <li>iii. NeuroFuzzy System</li> </ol>                                                                                                                                 | 10 |